

Tristan Louis

Queens, NY | 3474402592 | tristanlouie7@gmail.com | <https://www.linkedin.com/in/tristan-louis-097659236/> | <https://tristdismiss.github.io/>

EDUCATION

Queens College
Bachelor of Arts in Computer Science

Expected to Graduate: December 2026

EXPERIENCE

Data Science Intern

Dakdan Worldwide—Remote

August 2025 - Dec 2025

- Built automated ETL pipelines using Python and Pandas to ingest and transform sports-business datasets from Google Trends, Twitter, and Twitch
- Automated data validation and cleaning workflows, improving dataset consistency and reducing manual processing effort
- Developed analytical dashboards and reporting pipelines used to support marketing, sponsorship, and revenue-related decisions
- Optimized data transformation processes to improve reliability and maintainability of recurring reporting workflows

Software Development Intern

OpenQuantify—Remote

June 2025 - Dec 2025

- Developed backend services using FastAPI, Flask, SQLAlchemy, and PostgreSQL to support RESTful APIs and AI-powered application features
- Designed modular service architecture separating API, business logic, and data-access layers to improve scalability and maintainability
- Implemented database models and query optimization techniques to improve data retrieval performance
- Contributed to AI-driven simulation features using CesiumJS and integrated backend services supporting digital twin workflows

Data Engineer Fellow

Open Avenues Foundation: The Build Fellowship Highlight Reel: Automated Sports Content Generation

September 2024 – December 2024

- Built a machine learning pipeline using OpenCV and XGBoost to automate sports highlight generation from video footage
- Developed feature extraction and ball-tracking workflows to identify game events in real time
- Improved highlight detection accuracy from 67% to 89% through feature engineering and iterative model optimization
- Collaborated with engineers and industry mentors to deploy and evaluate end-to-end data processing workflows

PROJECTS

Real-Time Global News Intelligence Pipeline & Analytics Dashboard

Python, PySpark, PostgreSQL, FastAPI, Streamlit, Hugging Face (Transformers/BERT), JDBC, REST APIs, Git

- Built an end-to-end ETL pipeline that ingests global news data from multiple APIs and RSS feeds
- Implemented transformation and ranking logic using Pandas to generate country-specific Top 10 news insights
- Designed PostgreSQL data models and integrated SQLAlchemy ORM for efficient storage and retrieval
- Developed FastAPI endpoints and a Streamlit dashboard for interactive filtering, search, and analytics

PathBot Inventory Assistant

Python, FastAPI, PostgreSQL, SQLAlchemy, EasyOCR

- Architected a high-throughput spatial search API using FastAPI and PostgreSQL, migrating legacy flat-file JSON storage to a relational schema with B-Tree indexes to scale inventory lookups
- Engineered an L1 in-memory application cache with TTL invalidation, reducing database query load and achieving sub-5ms response times for high-frequency natural language search queries
- Developed a multi-modal ingestion engine integrating EasyOCR computer vision and SpeechRecognition pipelines to parse unstructured voice audio and handwritten shopping lists into structured SQL lookups
- Automated a weekly ETL scraping pipeline using BeautifulSoup to parse promotional store data, upserting active sale items into PostgreSQL for dynamic rendering in a Streamlit operational dashboard

Predictive Analytics for Diabetes Management

Python, Scikit-Learn, AWS SageMaker, Pandas, PCA, Apriori, Matplotlib

- Built machine learning models to predict medication adjustments and emergency healthcare utilization using patient treatment data
- Applied PCA and feature scaling techniques to improve model performance and training efficiency
- Used Apriori association rule mining to identify relationships between A1C levels, medication changes, and patient outcomes
- Deployed and evaluated models within AWS SageMaker to support scalable experimentation workflows

Customer Credit Risk Prediction using Machine Learning

Kaggle, Pandas, NumPy, scikit-learn (Logistic Regression, Decision Tree, Random Forest), XGBoost, Matplotlib, Seaborn, GridSearchCV, SMOTE

- Built a credit risk assessment model to predict loan default probabilities using XGBoost & Random Forest
- Engineered financial features like debt-to-income ratio, credit utilization, and payment history trends, improving accuracy to 88%
- Implemented SHAP (Shapley Additive Explanations) for explainability, providing financial analysts with insights into risk factors.
- Built an interactive dashboard using Streamlit, enabling lenders to assess borrower risk in real time

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Engineering:** ETL Pipelines, Data Modeling, PostgreSQL, SQLAlchemy, Pandas, NumPy
- **Backend:** FastAPI, Flask, REST APIs, SQLAlchemy
- **Machine Learning & AI:** XGBoost, Scikit-learn, OpenCV, TensorFlow, Feature Engineering,
- **Cloud & Deployment:** AWS (S3, Lambda, SageMaker), GCP

CERTIFICATES & TRAINING

- Build Fellowship Completion—An AI-powered sports highlight generation using ML & computer vision.
- Code Path Intermediate Technical Prep (TIP102)—Python, data structures, and algorithms